Assignment 9: Advanced SQL Queries Using Joins and Subqueries

DATABASE MANAGEMENT SYSTEM

ALLIANCE UNIVERSITY

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COURSE: BCA

YEAR: 2ND YEAR

PROJECT TITLE: Online Food Delivery System

1. Aim

To develop advanced SQL queries using joins and subqueries to retrieve related information, generate reports, and analyze data in the Online Food Delivery System.

2. Software Requirements

- Database Management System: MySQL
- Tool: MySQL Workbench
- Language: SQL

3. Theory

Joins combine records from two or more tables using related columns.

- **INNER JOIN:** Returns matching records from both tables.
- **LEFT JOIN:** Returns all records from the left table and matching records from the right table.
- **Subquery:** A query written inside another SQL query to retrieve data used by the outer query.

Joins and subqueries help generate reports about customers, orders, food items, payments, restaurants, and delivery partners.

4. Advanced SQL Queries with Outputs and Explanations

The following queries use the sample data inserted in Assignment 7.

4.1 Report on Customer Orders Using INNER JOIN

Input:

```
396 •  USE online_food_delivery;
397 •  SELECT
398        o.Order_ID,
399        c.Name AS Customer_Name,
400        o.Order_Date,
401        o.Total_Amount,
402        o.Order_Status
403    FROM Orders o
404    INNER JOIN Customer c
405        ON o.Customer_ID = c.Customer_ID
406    ORDER BY o.Order_ID;
```

Output:

Order_ID	Customer_Name	Order_Date	Total_Amount	Order_Status
10	Keerthana	2026-10-15 12:30:00	360.00	Delivered
20	Ravi	2026-10-15 13:00:00	120.00	Delivered
30	Priya	2026-10-16 19:15:00	250.00	Preparing
40	Arun	2026-10-17 20:00:00	300.00	Out for Delivery
50	Divya	2026-10-18 16:30:00	200.00	Placed

Explanation: This query joins the Orders and Customer tables to display each order with the customer's name, order date, amount, and status.

4.2 Report on Food Items and Restaurants Using Multiple Joins

Input:

```
408 •   SELECT
409         f.Food_Name,
410         r.Restaurant_Name,
411         c.Category_Name,
412         f.Price
413     FROM Food_Item f
414     INNER JOIN Restaurant r
415         ON f.Restaurant_ID = r.Restaurant_ID
416     INNER JOIN Category c
417         ON f.Category_ID = c.Category_ID
418     ORDER BY r.Restaurant_Name, f.Food_Name;
```

Output:

Food_Name	Restaurant_Name	Category_Name	Price
Veg Burger	Burger Hub	Burger	120.00
Farmhouse Pizza	Pizza Corner	Pizza	320.00
Margherita Pizza	Pizza Corner	Pizza	250.00
South Indian Meals	South Indian Meals	Indian	150.00
Paneer Butter Masala	Spice Garden	Indian	220.00
Veg Biryani	Spice Garden	Indian	180.00
Veg Noodles	Spice Garden	Chinese	160.00
Chocolate Cake	Sweet Treats	Desserts	100.00

Explanation: This query joins three tables to show each food item with its restaurant, category, and price.

4.3 Report on Orders, Food Items, and Quantities

Input:

```
420 •   SELECT
421         o.Order_ID,
422         c.Name AS Customer_Name,
423         f.Food_Name,
424         oi.Quantity,
425         oi.Subtotal
426     FROM Orders o
427     INNER JOIN Customer c
428         ON o.Customer_ID = c.Customer_ID
429     INNER JOIN Order_Item oi
430         ON o.Order_ID = oi.Order_ID
431     INNER JOIN Food_Item f
432         ON oi.Food_ID = f.Food_ID
433     ORDER BY o.Order_ID;
```

Output:

Order_ID	Customer_Name	Food_Name	Quantity	Subtotal
10	Keerthana	Veg Biryani	2	360.00
10	Keerthana	Veg Biryani	2	360.00
20	Ravi	Veg Burger	1	120.00
20	Ravi	Veg Burger	1	120.00
30	Priya	Margherita Pizza	1	250.00
30	Priya	Margherita Pizza	1	250.00
40	Arun	South Indian Meals	2	300.00
40	Arun	South Indian Meals	2	300.00
50	Divya	Chocolate Cake	2	200.00
50	Divya	Chocolate Cake	2	200.00

Explanation: This query joins four tables to display the customer, ordered food item, quantity, and subtotal for every order.

4.4 Report on Orders and Delivery Partners Using LEFT JOIN

Input:

```
435 •   SELECT
436         o.Order_ID,
437         c.Name AS Customer_Name,
438         dp.Partner_Name,
439         d.Delivery_Status
440     FROM Orders o
441     INNER JOIN Customer c
442         ON o.Customer_ID = c.Customer_ID
443     LEFT JOIN Delivery d
444         ON o.Order_ID = d.Order_ID
445     LEFT JOIN Delivery_Partner dp
446         ON d.Partner_ID = dp.Partner_ID
447     ORDER BY o.Order_ID;
```

Output:

Order_ID	Customer_Name	Partner_Name	Delivery_Status
10	Keerthana	Karthik	Delivered
20	Ravi	Suresh	Delivered
30	Priya	Manoj	Preparing
40	Arun	Vijay	In Transit
50	Divya	Naveen	Assigned

Explanation: This query shows each order with its assigned delivery partner and delivery status. LEFT JOIN ensures that orders are included even when delivery details are not yet available.

4.5 Subquery to Find Food Items Above the Average Price

Input:

```
449 •   SELECT Food_Name, Price
450     FROM Food_Item
451     WHERE Price > (
452         SELECT AVG(Price)
453         FROM Food_Item
454     )
455     ORDER BY Price DESC;
```

Output:

	Food_Name	Price
▶	Farmhouse Pizza	320.00
	Margherita Pizza	250.00
	Paneer Butter Masala	220.00

Explanation: The inner query calculates the average food price of ₹187.50. The outer query displays food items priced above that average.

4.6 Subquery to Find Customers Who Placed Orders Above the Average Order Amount

Input:

```
457 •   SELECT Name AS Customer_Name
458     FROM Customer
459     WHERE Customer_ID IN (
460         SELECT Customer_ID
461         FROM Orders
462         WHERE Total_Amount > (
463             SELECT AVG(Total_Amount)
464             FROM Orders
465         )
466     );
```

Output:

	Customer_Name
▶	Keerthana
	Priya
	Arun

Explanation: The average order amount is ₹246.00. This query finds customers whose orders exceed that average.

4.7 Summary Report of Orders by Status

Input:

```
457 •   SELECT Name AS Customer_Name
458     FROM Customer
459     WHERE Customer_ID IN (
460         SELECT Customer_ID
461         FROM Orders
462         WHERE Total_Amount > (
463             SELECT AVG(Total_Amount)
464             FROM Orders
465         )
466     );
```

Output:

Order_Status	Total_Orders	Total_Amount
Delivered	2	480.00
Out for Delivery	1	300.00
Placed	1	200.00
Preparing	1	250.00

Explanation: This query groups orders by status and calculates the number of orders and their combined amount for each status.

4.8 Subquery to Find Restaurants Offering Food Above ₹200

Input:

```
468 •   SELECT Restaurant_Name
469     FROM Restaurant
470     WHERE Restaurant_ID IN (
471         SELECT Restaurant_ID
472         FROM Food_Item
473         WHERE Price > 200
474     )
475     ORDER BY Restaurant_Name;
```

Output:

Restaurant_Name
Pizza Corner
Spice Garden

Explanation: The subquery finds restaurant IDs associated with food items priced above ₹200. The outer query displays the corresponding restaurant names.

4.9 Payment Report Using JOIN

Input:

```
468 •   SELECT
469         o.Order_ID,
470         c.Name AS Customer_Name,
471         p.Payment_Method,
472         p.Amount,
473         p.Payment_Status
474     FROM Orders o
475     INNER JOIN Customer c
476         ON o.Customer_ID = c.Customer_ID
477     INNER JOIN Payment p
478         ON o.Order_ID = p.Order_ID
479     ORDER BY o.Order_ID;
```

Output:

Order_ID	Customer_Name	Payment_Method	Amount	Payment_Status
10	Keerthana	UPI	360.00	Completed
20	Ravi	Card	120.00	Completed
30	Priya	UPI	250.00	Pending
40	Arun	Cash on Delivery	300.00	Pending
50	Divya	Card	200.00	Completed

Explanation: This query joins the order, customer, and payment tables to show payment methods, amounts, and payment statuses for each customer order.

5. Procedure

1. Open MySQL Workbench and connect to the database server.
2. Select the Online_Food_Delivery database.
3. Ensure that the sample data from Assignment 7 is available.
4. Execute each SQL query separately.
5. Compare the actual results with the expected outputs.
6. Capture screenshots of the SQL statements and result grids for submission.

6. Result

Advanced SQL queries were developed using INNER JOIN, LEFT JOIN, and subqueries. Reports were generated for customer orders, restaurant menus, ordered food items, delivery partners, payment details, and order summaries.

7. Conclusion

Joins and subqueries help retrieve related information from multiple database tables and perform useful analysis. These queries provide meaningful reports that support order tracking, payment monitoring, food price analysis, and delivery management in the Online Food Delivery System.